Generally, SSTP is used for remote access, i.e., by an employee accessing a network remotely from a different office or home who has no support for network-to-network tunnels. Performance wise, the quality of SSTP will degrade if the bandwidth is limited, and any lack of bandwidth can lead to sessions expiring.

- *Multi-Protocol Label Switching (MPLS):* This is used within the computer network infrastructure to speed up the data flow from one point to another. It implements and uses labels for routing decisions. MPLS operates by assigning a unique label or identifier to each network packet. The label consists of the routing table information, such as the destination IP address, bandwidth and other factors, as well as the source IP and socket information. The router can refer only to the label to make the routing decision rather than look into the packet. MPLS supports IP, asynchronous transfer mode (ATM), frame relay, synchronous optical networking (SONET) and Ethernet based networks. MPLS is designed to be used on both packet-switched networks and circuit-switched networks.
- *Hybrid approach:* Modern VPN networks use a blend of MPLS and IPSec protocols to leverage broadband bonding routers to make a cost-effective and reliable VPN architecture.

So, considering the adaptability of VPN, lots of VPN services have been created to address the increasing need to create custom VPN connections.

In this article, we list the top open source tools available for sysadmins to create their own VPNs.

SoftEther

SoftEther (Software Ethernet) is an open source, cross-platform, multi-protocol VPN client and VPN server software that supports almost all VPN protocols like SSL VPN, L2TP/IPSec, OpenVPN and Microsoft SSTP protocol. It provides sysadmins with a strong alternative to other VPN products like OpenVPN, IPSec and MS-SSTP, and is considered to have the best throughput, low latency and high resistance to firewalls. SoftEther VPN optimises performance by full Ethernet frame utilisation, reducing memory copy operations, parallel transmission and clustering.

Components

- *SoftEther VPN server:* This implements the VPN server, and listens to and accepts connections coming from VPN clients or the VPN bridge with varied VPN protocols.
- *SoftEther VPN client:* This has a virtualised function of the Ethernet network adapter. It has advanced functions like the best VPN communication settings compared to other VPN clients.
- *SoftEther VPN bridge:* This builds site-to-site VPNs, and systems administrators use it to install the SoftEther VPN server on the central server and SoftEther VPN bridge on remote sites.
- *VPN server manager:* This is a GUI based tool for systems administrators for the SoftEther VPN server and bridge.
- *Command-line admin utility:* This program runs on the consoles of every supported operating system. When users are unable to use Windows or Linux with Wine, they can alternatively use *vpncmd* to manage the VPN programs. *vpncmd* is also useful to execute a batch operation, such as creating many users on the virtual hub, or creating many virtual hubs on the VPN server.

Features

- Supports site-to-site and the remote access VPN with AES-256 and 4096-bit encryption.
- High speed throughput performance up to 1Gbps; IPv4 / IPv6 dual-stack; no memory leaks.
- Supports all VPN protocols like OpenVPN, IPSec, L2TP, MS-SSTP, EtherIP, L2TPv3; multi-language support and deep-inspect packet logging function.
- Cross-platform support including for Windows, Mac, Android, iOS, Linux, etc.

Official website: *https://www.softether.org/*
Latest version: 4.28

Algo VPN

Algo VPN is open source software that's specially designed for self-hosted IPSec VPN services. It was designed by the folks at Trail of Bits for fast deployment, using modern protocols and ciphers to enhance security. It comprises a set of Ansible scripts for easy setup by systems admins for creating personal IPSec based VPNs. Algo automatically deploys an on-demand VPN service in the cloud that is not shared with other users.

Features

- Supports sysadmins with a helper script to add or remove users from the network instantly.
- Supports only IKEv2 with encryption algorithms, i.e., AES-GCM, SHA2, P-256 and Wireguard.
- Facilitates configuring limited SSH users for tunnelling traffic.
- Supports DigitalOcean, Microsoft Azure, Google Compute Engine, OpenStack, Vultr and Ubuntu 18.04 server.

Official website: *https://github.com/trailofbits/algo*

Streisand

Streisand only supports installation on the Ubuntu 16.04 server using single commands, and supports L2TP, OpenConnect, OpenSSH, OpenVPN, Shadowsocks, Stunnel, Tor Bridge and Wireguard. Depending on the protocol chosen by the user, apps can be installed. It has many features that match those of Algo, but provides more protocol support for sysadmins. Streisand is installed using Ansible, and users can be added easily using custom-generated connection instructions. It includes an embedded copy of the server's SSL certificate.